MATH INQUIRERS HOLIDAY PROGRAM

Weekly Email Pack

Parent and tutor emails for the five weeks of the P5 Ages 11-12 holiday program

Notice it. Test it. Explain why.

How to use this pack

- Send the parent email and the Week N Student Pages PDF the day before the first lesson of that week.
- Send the tutor email with the Week N Tutor Scripts and Answers PDF at least two days before. Tutors only; the answers must not go to parents.
- Put parents in BCC so no one sees another family's email address. Get parents' consent to email before you start (PDPA).
- Replace the words in [square brackets] before sending.
- The PDFs are in the weekly folder (Math_Inquirers_P5_Weekly_PDFs.zip): Week1_Student_Pages.pdf to Week5_Student_Pages.pdf and Week1_Tutor_Scripts_and_Answers.pdf to Week5_Tutor_Scripts_and_Answers.pdf. The Week 1 student PDF also holds the inquiry journal page and all the cut-outs; the Week 5 PDF ends with the certificate.
- The slide deck links in the tutor emails are private to you. Share each deck with the tutor first, or send them the file.

Week 1: Lessons 1 and 2

Parent email

Subject: Math Inquirers P5 Ages 11-12 Week 1: Numbers to 10 Million and Fractions: Four Operations

Attachment: Week1_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 11-12 (finishing P5)) meet two topics, in lessons 1 and 2.

Lesson 1, Million Puzzles (Numbers to 10 Million). Reading, comparing and rounding numbers up to 10 million, and riddles about large numbers.

Lesson 2, Fraction Feats (Fractions: Four Operations). Adding, subtracting, multiplying and dividing fractions, using bar models and pictures.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

The cut-out pages are at the end of the Week 1 PDF (Fraction wall, square grid, dot grid, cube net). Please cut them out and keep them in one envelope; we use them through the program. The inquiry journal page is at the front. This week's lessons use: fraction wall.

One thing to try at home (5 minutes): Find a big number in the news, such as a population or a budget. Ask your child to say it in words and round it to the nearest million.

Ask your child: "Which digit decides how you round?" or "Can you show it with a bar?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 1: Tutor email

Subject: Math Inquirers P5 Ages 11-12 Week 1 tutor pack: Lessons 1 and 2

Attachments: Week1_Tutor_Scripts_and_Answers.pdf, Week1_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 1: Million Puzzles; Fraction Feats.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 1 deck: https://claude.ai/artifact/CpsbcBG7fSWGLD4CUnayrU
- Lesson 2 deck: https://claude.ai/artifact/M7yiFV8xGfVMM1RKDQebb4

Before the lesson, please:

1. Read both scripts once and try the Challenge questions yourself.
2. Check your materials are ready. Lesson 1: Digit cards 0 to 9 (one set per child), a 7-column place-value mat (M, HTh, TTh, Th, H, T, O), mini whiteboard, a calculator for the explore only, workbook. Lesson 2: Paper circles (6 per child) to cut as pizzas, scissors, a strip of paper for bar models, digit cards 1 to 4, mini whiteboard, workbook.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 2: Lessons 3 and 4

Parent email

Subject: Math Inquirers P5 Ages 11-12 Week 2: Fraction Word Problems and Area of Triangle

Attachment: Week2_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 11-12 (finishing P5)) meet two topics, in lessons 3 and 4.

Lesson 3, Before and After (Fraction Word Problems). Fraction word problems: before-and-after situations and 'fraction of the remainder', solved with bar models.

Lesson 4, Triangle Areas (Area of Triangle). Area of a triangle: base and height, why the area is half of a rectangle, and triangles in composite figures.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: fraction wall, square grid.

One thing to try at home (5 minutes): Make up a story: "I spent 1/4 of my money, then 1/2 of the rest, and had $12 left." Let your child draw it and work backwards.

Ask your child: "What is the whole here, and what is it now?" or "Which side is the base, and where is the height?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 2: Tutor email

Subject: Math Inquirers P5 Ages 11-12 Week 2 tutor pack: Lessons 3 and 4

Attachments: Week2_Tutor_Scripts_and_Answers.pdf, Week2_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 2: Before and After; Triangle Areas.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 3 deck: https://claude.ai/artifact/4q3zF6YdSAEbJW21ArCMFW
- Lesson 4 deck: https://claude.ai/artifact/SETLyEwKG99PMqJbS4CVgF

Before the lesson, please:

1. Read both scripts once and try the Challenge questions yourself.
2. Check your materials are ready. Lesson 3: Mini whiteboard and marker, 24 counters or paper squares (two colours), paper strips for bar models, workbook, pencil. Lesson 4: Square grid paper (cut-out page at the back of the workbook), scissors, ruler, set square or the corner of a card, mini whiteboard and marker, workbook, pencil.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 3: Lessons 5 and 6

Parent email

Subject: Math Inquirers P5 Ages 11-12 Week 3: Volume and Rate

Attachment: Week3_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 11-12 (finishing P5)) meet two topics, in lessons 5 and 6.

Lesson 5, Volume Vault (Volume). Volume of cuboids, water in tanks, stones that sink, and painted-cube puzzles.

Lesson 6, Rate Racers (Rate). Rate: the amount for one, better buys, taps filling a tank, and working out a charge 'or part of' an hour.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: cube net.

One thing to try at home (5 minutes): Fill a box with sugar cubes or blocks. Ask for the number of cubes before counting, then check.

Ask your child: "How many cubes in one layer, and how many layers?" or "What does it cost for just one?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 3: Tutor email

Subject: Math Inquirers P5 Ages 11-12 Week 3 tutor pack: Lessons 5 and 6

Attachments: Week3_Tutor_Scripts_and_Answers.pdf, Week3_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 3: Volume Vault; Rate Racers.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 5 deck: https://claude.ai/artifact/3BjB4bmWiTJ6o8tAq9qCks
- Lesson 6 deck: https://claude.ai/artifact/72QFCVitzrebZbPhUbBoeG

Before the lesson, please:

1. Read both scripts once and try the Challenge questions yourself.
2. Check your materials are ready. Lesson 5: About 30 one-centimetre cubes (or dice, Lego bricks, sugar cubes), a ruler, squared paper, mini whiteboards. Tutor: a clear plastic box with a ruler taped to its side, water and a stone for the demonstration. Lesson 6: Mini whiteboards, a calculator for checking only, a stopwatch (or phone), a few real price tags or supermarket flyers if you have them.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 4: Lessons 7 and 8

Parent email

Subject: Math Inquirers P5 Ages 11-12 Week 4: Percentage and Angles, Triangles and Quadrilaterals

Attachment: Week4_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 11-12 (finishing P5)) meet two topics, in lessons 7 and 8.

Lesson 7, Percentage Puzzles (Percentage). Percentage as a fraction of 100, finding a part, discount, GST and percentage change.

Lesson 8, Angle Chasers (Angles, Triangles and Quadrilaterals). Angle rules for triangles and quadrilaterals, finding unknown angles and giving a reason for each step.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

Please keep the Week 1 envelope handy. This week's lessons use: dot grid.

One thing to try at home (5 minutes): Look at a sale tag. Ask for the price after 20% off, then check with a calculator.

Ask your child: "Percentage of what? What is the 100%?" or "Which angle rule did you use, and why?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 4: Tutor email

Subject: Math Inquirers P5 Ages 11-12 Week 4 tutor pack: Lessons 7 and 8

Attachments: Week4_Tutor_Scripts_and_Answers.pdf, Week4_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 4: Percentage Puzzles; Angle Chasers.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 7 deck: https://claude.ai/artifact/FUuPskuuqaDx2ZVzpt5u5u
- Lesson 8 deck: https://claude.ai/artifact/8kbz6bXBusNwpuycR9RfZk

Before the lesson, please:

1. Read both scripts once and try the Challenge questions yourself.
2. Check your materials are ready. Lesson 7: Mini whiteboard and marker, workbook, pencil, a hundred-square page (or squared paper), a calculator for checking only at the end of each page. Lesson 8: Paper, ruler, pencil, scissors, protractor, red and blue pencils, the dot-grid cut-out, mini whiteboards, workbook.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5: Lessons 9 and 10

Parent email

Subject: Math Inquirers P5 Ages 11-12 Week 5: Patterns and Number Sense and Mixed Heuristics

Attachment: Week5_Student_Pages.pdf

Dear parents,

This week our Math Inquirers (Ages 11-12 (finishing P5)) meet two topics, in lessons 9 and 10.

Lesson 9, Pattern Generalisers (Patterns and Number Sense). Patterns and number sense: finding the rule, predicting far-away terms, and checking with a table.

Lesson 10, Challenge Day (Mixed Heuristics). Choosing a strategy: guess and check, working backwards, making a table and drawing a model.

Your child's workbook pages for this week are attached. Please print them single-sided if you can. Children will work on the pages in class, so there is no need to finish them in advance.

No cut-outs are needed this week. Please keep the Week 1 envelope handy. The Certificate of Completion is the last page of the Week 5 PDF.

One thing to try at home (5 minutes): Make a matchstick pattern. Ask for the number of sticks in the 20th figure before you build it.

Ask your child: "Can you test your rule on the next figure?" or "How did you start? What would you try next time?"

We are not looking for speed. We are looking for a child who can say what they noticed, test it, and explain why. If your child is stuck, please ask "What do you notice?" before you give the answer.

Warm regards,
[Your name]
Preciouslyours Pte Ltd
[Contact number]

Week 5: Tutor email

Subject: Math Inquirers P5 Ages 11-12 Week 5 tutor pack: Lessons 9 and 10

Attachments: Week5_Tutor_Scripts_and_Answers.pdf, Week5_Student_Pages.pdf

Hi [tutor name],

Here is your pack for week 5: Pattern Generalisers; Challenge Day.

Attached are the minute-by-minute scripts with answers for both lessons, and the student pages. The slide decks are linked below.

- Lesson 9 deck: [deck link to add: not yet published]
- Lesson 10 deck: [deck link to add: not yet published]

Before the lesson, please:

1. Read both scripts once and try the Challenge questions yourself.
2. Check your materials are ready. Lesson 9: Matchsticks or toothpicks (about 20 per child, 40 if possible), square paper tiles or the square grid from the cut-out pages, mini whiteboard, workbook. Lesson 10: Four puzzle cards shown on the slide, or sent as images to each breakout room (backwards, logic, counting, guess and check), mini whiteboards, scrap paper, a 3 by 4 grid drawn on paper, certificates, stamps.
3. After each lesson, tick the Readiness Tracker for each child (four quick ticks) and send me the sheet.

Please do not forward the answer pages to parents.

Thank you,
[Your name]

Week 5 closing email to parents

Subject: Math Inquirers P5 Ages 11-12: thank you, and your child's certificate

Dear parents,

Thank you for joining us for the Math Inquirers Holiday Program (Ages 11-12 (finishing P5)). Your child's Certificate of Completion is the last page of the Week 5 PDF. Please print it and add your child's name.

We would love your feedback in two minutes: [feedback form link]. What did your child enjoy most? What should we change?

If your child showed a love for puzzles, we may invite them to a short challenge session in the next holiday. We will contact you if so.

Warm regards,
[Your name]